ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 F ✓ 1.3.2 E ✓ 1.3.3 J ✓ 1.3.4 D ✓ 1.3.5 G ✓ 1.3.6 C ✓ 1.3.7 I ✓ 1.3.8 H ✓ 1.3.9 B ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FOUR: always put the hot iron back in its stand ✓; work in a well-ventilated area / use a fume extractor ✓; wear safety glasses ✓; never touch the tip ✓; switch off and unplug after use; wash hands after handling solder; keep the cable away from the hot tip (4)

2.2 Current – the flow of electric charge (electrons) ✓, ampere (A) ✓; voltage – the potential difference / electrical pressure that pushes current ✓, volt (V) ✓; resistance – opposition to the flow of current ✓, ohm (Ω) ✓ (6)

2.3 The current through a conductor is directly proportional to the voltage across it at constant temperature ✓; V = I × R ✓ (2)

2.4 $V_R = 5 - 2 = 3\ V$ ✓; $R = V / I$ ✓ $= 3 / 0{,}02$ ✓ $= 150\ \Omega$ ✓ (4)

2.5 (a) $P = V \times I = 12 \times 1{,}5$ ✓ $= 18\ W$ ✓ (b) $R = V / I = 12 / 1{,}5$ ✓ $= 8\ \Omega$ ✓ (4)

[20]

QUESTION 3

3.1 (a) $R_T = 100 + 220 + 330$ ✓ $= 650\ \Omega$ ✓ (b) $I = 9 / 650$ ✓ $= 0{,}014\ A$ (13,85 mA) ✓ (c) $V = 0{,}01385 \times 220$ ✓ $= 3{,}05\ V$ ✓ (6)

3.2 $R_T = (60 \times 30) / (60 + 30)$ ✓ $= 1\,800 / 90$ ✓ $= 20\ \Omega$ ✓ (3)

3.3 Correct standard symbols: (a) NPN transistor – circle with base, collector and emitter, arrow pointing out on the emitter ✓ (b) motor – circle with M ✓ (c) push-button – contacts with a button bridge, normally open ✓ (3)

3.4 When the transistor switches off, the coil produces a high reverse voltage (back EMF) ✓; the diode (flyback diode) gives this current a safe path ✓ and protects the transistor / microcontroller from damage ✓. (3)

[15]

QUESTION 4

4.1 $(a)\ 45 = 32 + 8 + 4 + 1$ ✓ $\rightarrow 101101$ ✓ $(b)\ 128 + 64 + 32 + 4 + 1$ ✓ $= 229$ ✓ $(c)\ 1010 = A, 1111 = F$ ✓ $\rightarrow AF$ ✓ (6)

4.2 $107 + 29 = 136$ ✓ $\rightarrow 1000\ 1000$ ✓ (2)

4.3 $2^8 - 1$ ✓ $= 255$ ✓ $(1111\ 1111)$ (2)

[10]

QUESTION 5

5.1 Symbol: AND shape with a small circle (bubble) on the output ✓; truth table ✓✓✓✓: 0 0 → 1; 0 1 → 1; 1 0 → 1; 1 1 → 0 (5)

5.2.1 Gate 1: AND ✓; gate 2: NOT (inverter) ✓; gate 3: OR ✓ (3)

5.2.2 $X = A \cdot B + \bar{C}$ ✓✓ $(X = (A\ AND\ B)\ OR\ (NOT\ C))$ (2)

5.2.3 Truth table (1 mark per correct row):

A	B	C	D = A·B	E = C̄	X
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	1	0	1

(8)

5.2.4 No ✓ – X = 0, because only one bump sensor is pressed (D = 0) and the safety key is in place (E = 0) ✓. (2)

5.3 Y = G · (S1 + S2) ✓✓. Circuit: S1 and S2 into an OR gate ✓; OR output and G into an AND gate ✓; output labelled Y ✓ (5)

[25]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE